HENRIQUE K. MIYAMOTO

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CentraleSupélec, Université Paris-Salcay
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EDUCATION

Université Paris-Saclay, France — 2022–present
Ph.D. in Information Theory
Graduate School of Computer Science

Universidade Estadual de Campinas, Brazil — 2021–2022
M.Sc. in Applied Mathematics
Dissertation: Geometry, Statistics and Applications to Communications and Learning

CentraleSupélec, France — 2018–2021
Diplôme d'Ingénieur (M.Sc.)
Major: Applied Mathematics

Universidade Estadual de Campinas, Brazil — 2015–2021
B.Sc. in Electrical Engineering
Class rank: 1/74, GPA: 0.9196/1.0

RESEARCH INTERESTS

Information theory and communication theory (MSC2020 94A and 94B).

RESEARCH EXPERIENCE

2022–present, *doctoral researcher*, **L2S, CNRS/CentraleSupélec/Université Paris-Saclay**, France.

- Advisor: Prof. Sheng Yang.

2021–2022, *master's researcher*, **IMECC, Universidade Estadual de Campinas**, Brazil.

- Advisor: Prof. Sueli I. R. Costa.

2020–2020, *researcher intern*, **Huawei Technologies**, France.

- Supervisors: Dr. Ingmar Land, Dr. Apostolos Destounis and Prof. Jean-Claude Belfiore.

2019–2020, *student researcher*, **L2S, CNRS/CentraleSupélec/Université Paris-Saclay**, France.

- Supervisor: Prof. Sheng Yang.

2016–2018, *undergraduate student researcher*, **IMECC, Universidade Estadual de Campinas**, Brazil.

- Supervisors: Prof. Henrique N. Sá Earp and Prof. Sueli I. R. Costa.

TEACHING TRAINING

Université Paris-Saclay, France — *2022–present*
Certificate "Teaching in Higher Education" (*Label "Enseigner dans le Supérieur"*)

- This program requires training in different pedagogical skills and teaching experience of at least 96h, done in parallel to the preparation of the Ph.D. degree.

TEACHING EXPERIENCE

2022–present, *graduate teaching assistant*, **CentraleSupélec**, France.

- 1EL2500 – Information theory
- 1CC4000 – Signal processing
- 2EL2610 – Communication theory
- 3SQ1025 – Digital signal processing
- 3SQ2120 – Wireless communications

2022–2022, *graduate teaching assistant*, **Universidade Estadual de Campinas**, Brazil.

- EE881 – Principles of communications I

2021–2021, *undergraduate teaching assistant*, **Universidade Estadual de Campinas**, Brazil.

- EE881 – Principles of communications I

PREPRINTS

[Z1] H. K. Miyamoto and S. Yang, "Universal decoding over finite-state additive channels via noise guessing", *arXiv:2501.12971*, 2025. doi: 10.48550/arXiv.2501.12971.

JOURNAL PAPERS

[J4] H. K. Miyamoto, F. C. C. Meneghetti, J. Pinele and S. I. R. Costa, "On closed-form expressions for the Fisher–Rao distance", *Information Geometry*, vol. 7, pp. 311–354, Nov. 2024, doi: 10.1007/s41884-024-00143-2.

[J3] H. K. Miyamoto, F. C. C. Meneghetti and S. I. R. Costa, "The Fisher–Rao loss for learning under label noise", *Information Geometry*, vol. 6, pp. 107–126, Jun. 2023, doi: 10.1007/s41884-022-00076-8.

[J2] H. K. Miyamoto and S. Yang, "Context-tree-based lossy compression and its application to CSI representation", *IEEE Transactions on Communications*, vol. 70, no. 7, pp. 4417–4428, July 2022, doi: 10.1109/TCOMM.2022.3173002.

[J1] H. K. Miyamoto, S. I. R. Costa and H. N. Sá Earp, "Constructive spherical codes by Hopf foliations", *IEEE Transactions on Information Theory*, vol. 67, no. 12, pp. 7925–7939, Dec. 2021, doi: 10.1109/TIT.2021.3114094.

CONFERENCE PAPERS

[C9] H. K. Miyamoto and S. Yang, "On universal decoding over discrete additive channels by noise guessing", accepted to *IEEE Information Theory Workshop (ITW)*, Sydney, 2025.

[C8] H. K. Miyamoto and S. Yang "On universal decoding over memoryless channels with the Krichevsky–Trofimov estimator", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1498–1503, Athens, 2024, doi: 10.1109/ISIT57864.2024.10619414.

[C7] H. K. Miyamoto, "Geometria, estatística e aplicações a comunicações e aprendizado"[1], *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*, vol. 10, no. 1, 2023, doi: 10.5540/03.2023.010.01.0004 (in Portuguese).

[C6] F. C. C. Meneghetti, H. K. Miyamoto, S. I. R. Costa and M. H. M. Costa, "Revisiting lattice tiling decomposition and dithered quantisation", *Geometric Science of Information (GSI)*, Saint-Malo, 2023. In: F. Nielsen and F. Barbaresco (eds.), *GSI 2023*, LNCS 14071, pp. 318–327, Springer, 2023, doi: 10.1007/978-3-031-38271-0_319.

[1] This is an extended abstract of [T1], published on the occasion of the *Clóvis Caesar Gonzaga Award* (see Awards).

[C5] F. C. C. Meneghetti, H. K. Miyamoto and S. I. R. Costa, "Information properties of a random variable decomposition through lattices", *41st International Conference on Bayesian and Maximum Entropy Methods in Science and Engineering (MaxEnt)*, Paris, 2022. In: *Physical Sciences Forum*, vol. 5, no. 1, 2022, doi: 10.3390/psf2022005019.

[C4] H. K. Miyamoto and S. Yang, "A CSI compression scheme using context trees", *International Zurich Seminar on Information and Communication (IZS)*, pp. 24–28, Zurich, 2022, doi: 10.3929/ethz-b-000535273.

[C3] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Constructive spherical codes in 2^k dimensions", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1612–1616, Paris, 2019, doi: 10.1109/ISIT.2019.8849464.

[C2] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Construção de códigos esféricos usando a fibração de Hopf", *Jornada Nacional de Iniciação Científica (JNIC), 70ª Reunião Anual da SBPC*, Maceió, 2018, ISSN: 2176-1221 (in Portuguese).

[C1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Construction of spherical codes using the Hopf fibration", *XXV Congresso de Iniciação Científica da Unicamp*, Campinas, 2017, doi: 10.19146/pibic-2017-78809.

PATENT APPLICATIONS

[P1] I. Land, H. K. Miyamoto, A. Destounis and J.-C. Belfiore, "Transfer Learning between Neural Networks", WO2022207097 (2022), CN117121020 (2023), EP4309079 (2024).

THESES

[T1] H. K. Miyamoto, "Geometria, Estatística e Aplicações a Comunicações e Aprendizado" (Geometry, Statistics and Applications to Communications and Learning), *M.Sc. dissertation in applied mathematics*, Universidade Estadual de Campinas, Campinas, 2022 (in Portuguese).

PRESENTATIONS

IEEE International Symposium on Information Theory (ISIT), *oral presentation*, Athens, Greece, 2024.

IEEE European School on Information Theory (ESIT), *poster presentation*, Eindhoven, Netherlands, 2024.

France PhD Workshop on Information Theory, *poster presentation*, Palaiseau, France, 2024.

Geometric Science of Information (GSI), *oral presentation*, Saint-Malo, France, 2023.

IEEE European School on Information Theory (ESIT), *poster presentation*, Bristol, UK, 2023.

International Conference on Information Geometry for Data Science (IG4DS), *oral presentation*, Hamburg, Germany (virtual), 2022.

IEEE International Symposium on Information Theory (ISIT), *oral presentation*, Paris, France, 2019.

International Congress of Mathematicians (ICM), *oral presentation*, Rio de Janeiro, Brazil, 2018.

Jornada Nacional de Iniciação Científica (JNIC), *poster presentation*, Maceió, Brazil, 2018.

Latin American Week on Coding and Information (LAWCI), *poster presentation*, Campinas, Brazil, 2018.

Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), *oral presentation*, Curitiba, Brazil, 2017.

XXV Congresso de Iniciação Científica da Unicamp, *poster presentation*, Campinas, Brazil, 2017.

AWARDS

1st place in **Clóvis Caesar Gonzaga Award** for master's dissertations in applied and computational mathematics, SBMAC, 2023.

Merit Award in Jornada Nacional de Iniciação Científica (JNIC), SBPC, 2018.

2nd place in **Beatriz Neves Award** for undergraduate research in applied and computational mathematics, SBMAC, 2018.

Academic Excellence in Undergraduate Work in Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), category Geometry and Topology, PET Matemática/UFPR, 2017.

Scientific Merit in XXV Congresso de Iniciação Científica da Unicamp, PRP/Unicamp, 2017.

Merit Award for the composition produced in Unicamp entrance exam , COMVEST/Unicamp, 2015.

SCHOLARSHIPS

2021–2022, *Master's Degree Scholarship*, São Paulo Research Foundation (FAPESP), 2021/04516-8.

2018–2020, *Eiffel Excellence Scholarship*, French Ministry of Europe and Foreign Affairs.

2016–2018, *Scientific Initiation Scholarship*, São Paulo Research Foundation (FAPESP), 2016/05126-0.

2013–2013, *Junior Scientific Initiation Scholarship*, Brazilian National Council for Scientific and Technological Development (CNPq).

ACADEMIC SERVICE

Scientific societies membership

2020–present, *student member*, Brazilian Society for the Advancement of Science (SBPC).

2018–present, *graduate student member*, Institute of Electrical and Electronics Engineers (IEEE).

- Information Theory Society, Communications Society, Signal Processing Society.

Reviewing

2025, *reviewer*, Information Geometry.

2023–2025, *reviewer*, IEEE International Symposium on Information Theory (ISIT).

2023, *reviewer*, Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT).

2022, *reviewer*, Journal of Communication and Information Systems (JCIS).

Committees and representative positions

2023–present, *doctoral and postdoc researchers representative*, Laboratory of Signals and Systems (L2S).

2023, *organising committee*, Encontro de Códigos, Reticulados e Informação, IMECC, Unicamp.

2018–2019, *student representative*, Departamento de Comunicações, FEEC, Unicamp.

LANGUAGES

Portuguese	native
English	advanced (Cambridge CAE; TOEFL ITP: 660/677)
French	advanced (TFI: 975/990)
Spanish	intermediate
Japanese	elementary (JLPT N4)

Last updated: August 13, 2025